

Ramchander Rao Bhaskara

ramchander.rao.bhaskara@dartmouth.edu | [ram-bhaskara.github.io](https://github.com/ram-bhaskara)

Academic Appointments

Dartmouth College Department of Computer Science <i>Postdoctoral Associate</i> <i>Advised by Prof. Adithya Pediredla</i>	Dec 2025 – present
Texas A&M University Department of Aerospace Engineering <i>Research Engineer</i>	Sep – Dec 2025

Education

Texas A&M University PhD in Aerospace Engineering <i>Advised by Prof. Manoranjan Majji</i>	2021 – 2025
Texas A&M University M.S. in Aerospace Engineering <i>Advised by Prof. Manoranjan Majji</i>	2019 – 2021
National Institute of Technology, Trichy BTech in Instrumentation and Control Engineering	2013 – 2017

Awards

Future Leaders in Aerospace Symposium Smead Aerospace Engineering Sciences, CU Boulder <i>Invited participant</i>	2026
Best Talk Award  Postdoctoral Research Day, Dartmouth College and University of New Hampshire	2026
Excellent Reviewer Journal of Guidance, Control, and Dynamics (JGCD)	2025
Guidance, Navigation, and Control (GNC) Graduate Award  American Institute of Aeronautics and Astronautics <i>3,500 USD</i>	2024
Graduate Excellence Fellowship Texas A&M University <i>5 × 1,000 USD</i>	2020 – 2024
Travel Award Department of Aerospace Engineering, Texas A&M University <i>2 × 500 USD</i>	2023 – 2024
Finalist, GNC Graduate Student Paper Award AIAA SciTech Forum <i>500 USD</i>	2023
Best Student Research Paper Award  Digital Avionics Systems Conference (DASC) <i>500 USD</i>	2022
Travel Grant Office of Graduate and Professional Studies, Texas A&M University <i>500 USD</i>	2022
ASIE Scholarship American Society of Indian Engineers and Architects, Houston <i>1,000 USD</i>	2022
NASA TechLeap Prize — GNC Lead  NASA Flight Opportunities Program <i>100,000 USD towards flight experiment</i>	2021
RECT Silver 72 Scholarship National Institute of Technology, Trichy <i>15,000 INR per year</i>	2015 – 2017

IIT Madras Summer Research Fellowship

2016

Department of Aerospace Engineering, IIT Madras | *Granted but declined*

Other Research Experience

Cornell University

2025

Mechanical and Aerospace Engineering | *SmallSat Mission Design School*

NASA Jet Propulsion Laboratory

2022, 2023

Mobility and Robotic Systems Section | *Research Intern*

iRunway India (now UnitedLex)

2017 – 2019

Intellectual Property | *Associate (5G/LTE network architecture)*

Publications

SDR-ToF: Million Optical Depth Samples per Second Using Software-Defined Radio [🔗](#)

Ramchander Rao Bhaskara*, Dhawal Sirikonda*, Omkar Vengurlekar*, Juhyeon Kim, Joseph Lazarro, Suren Jayasuriya, Adithya Pediredla

ACM Transactions on Graphics (SIGGRAPH Asia), 2026

ShutterEvents: Shutter Modulation for Static-Dynamic Scene Recovery [🔗](#)

Nevindu Batagoda*, Ramchander Rao Bhaskara*, Christian Metzler, Adithya Pediredla

IEEE International Conference on Computational Photography (ICCP) 2026

Development and Validation of Velocimeter Lidar Simulator [🔗](#)

Ramchander Rao Bhaskara, Roshan T. Eapen, and Manoranjan Majji

AIAA SciTech 2025

Quantized State Estimation for Linear Dynamical Systems [🔗](#)

Ramchander Rao Bhaskara, Manoranjan Majji, and Felipe Guzmán

Sensors 2024

Icy Moon Surface Simulation and Stereo Depth Estimation for Sampling Autonomy [🔗](#)

Ramchander Rao Bhaskara, G Georgakis, J Nash, J Bowkett, M Cameron, A Ansar, P Backes, and M Majji

IEEE Aerospace Conference 2024

Differentiable Rendering for Pose Estimation in Proximity Operations [🔗](#)

Ramchander Rao Bhaskara, Roshan T. Eapen, and Manoranjan Majji

AIAA SciTech Forum 2023 | [Finalist, GNC graduate student papers](#)

An FPGA Framework for Interferometric Vision-Based Navigation (iVisNav) [🔗](#)

Ramchander Rao Bhaskara, Kookjin Sung, and Manoranjan Majji

41st Digital Avionics and Systems Conference 2022 | [Best student research paper](#)

FPGA Hardware Acceleration for Feature-Based Relative Navigation Applications [🔗](#)

Ramchander Rao Bhaskara and Manoranjan Majji

AAS/AIAA Astrodynamics Specialist Conference 2022

Vision and Inertial Sensor Fusion for Terrain Relative Navigation [🔗](#)

A. Verras, R. Eapen, A Simon, M Majji, Ramchander Rao Bhaskara, C. Restrepo, and R. Lovelace

AIAA SciTech Forum 2021

Interferometric Vision-Based Navigation Sensor for Autonomous Proximity Operation [🔗](#)

Kookjin Sung, Ramchander Rao Bhaskara, and Manoranjan Majji

39th Digital Avionics and Systems Conference 2020

Under Review

On the Low-Pass Filter Design for Digital Phasemeter

Ramchander Rao Bhaskara and Manoranjan Majji

Submitted to *IEEE Transactions on Instrumentation and Measurement* 2026

Patents

MHz-Rate Optical Depth Imaging Using Software-Defined Radios

Adithya Pediredla, Ramchander Rao Bhaskara, Dhawal Sirikonda, Omkar Vengurlekar, Juhyeon Kim, Joseph Lazzaro, Suren Jayasuriya

U.S. Patent Application No.: 64/113,861

Other Publications

Real-Time Signal Processing and State Estimation for Spaceflight Applications [↗](#)

Ramchander Rao Bhaskara

PhD Dissertation, supervised by Prof. Manoranjan Majji

Hardware Implementation of Navigation Filters for Automation of Dynamical Systems [↗](#)

Ramchander Rao Bhaskara

Master's Thesis, supervised by Prof. Manoranjan Majji

Searching for Ariel's Aliens with the CRISPI Mass Spectrometer SmallSat Flyby Mission [↗](#)

Zach Ulibarri, S Jamal, A Alex, L Arzoumanian, Ramchander Rao Bhaskara, A Chadwick, S Gayen et al.

American Geophysical Union 2025

Precision Depth Sensing: Space Science to Robotics

Ramchander Rao Bhaskara

Guarini Poster Session, Dartmouth College, 2026

Toward Development and Validation of Velocimeter LiDAR Simulator

Ramchander Rao Bhaskara and Manoranjan Majji

2024 AAS Guidance, Navigation and Control Conference., Breckenridge, Colorado

NaRPA: Navigation and Rendering Pipeline for Astronautics [↗](#)

Roshan T. Eapen*, Ramchander Rao Bhaskara*, and Manoranjan Majji

Poster, Lunar Surface Innovation Consortium 2021

Physics-Based Modeling of Selective Catalytic Reduction System and Reduced Order Dynamics [↗](#)

Ramchander Rao Bhaskara

Undergraduate Thesis, supervised by Prof. Umapathy Mangalanathan

Talks

Precision depth sensing: space science and robotics

Future Leaders in Aerospace Symposium | *Invited talk*

University of Colorado Boulder | 2026

Postdoctoral Research Day

Dartmouth College | 2026

On-Board Sensing and State Estimation for Opto-Mechanical Inertial Sensors

Starlink (SpaceX) | *hosted by Javier Vicens (GNC)*

Seattle | 2025

Computer Vision for Icy-World Exploration

Machine-Ground Interaction Consortium (MaGIC) | *hosted by Prof. Dan Negrut*

UW Madison | 2025

On Sensor Simulations for Planetary Exploration

Visual Computing Seminar | *Invited Talk*

Dartmouth College | 2025

Sampling Autonomy for Europa Lander

Jet Propulsion Laboratory | *Intern Talk*

Pasadena, CA | 2023

Study of Topology of Icy Moons

Jet Propulsion Laboratory | *Intern Talk*

Pasadena, CA | 2023

Texas A&M ScORE: Space Object Rendering Engine

Lunar Surface Innovation Consortium | *Symposium*

Johns Hopkins University | 2021

Academic Service

International Conference on Computational Photography (ICCP)

2027

Website chair

American Institute of Aeronautics and Astronautics (AIAA)

2024 – Present

Sensor Systems and Information Fusion (SSIF) | *Technical Committee*

AIAA SciTech Forum

2025

Guidance, Navigation, and Control (GNC-44) | *Session Chair*

AIAA SciTech Forum

2025

Sensor and Algorithm Analysis (SEN-02) | *Session Chair*

Hosted colloquium talks: Georgios Georgakis (JPL, 2026), Jimin Wu (Rice University, 2026)

Hosted lab visits/demos: Whitney Mason (DARPA), Xia Zhou (Columbia University), Daphna Weinshall (Hebrew University); Anthony Rizzo, Bryce Ferguson, Alberto Quattrini Li, Jonathan Chipman (Dartmouth); Andrew Morgan (RAI Institute). Demos at ICCP 2026, TEROS 2023-24.

Reviews

Journal of Astronautical Sciences (JAS) (3)

2024 – 26

IEEE Transactions on Computational Imaging (1)

2026

Journal of Guidance, Control, and Dynamics (JGCD) (1)

2025

IEEE Control Systems Letters (1)

2025

American Control Conference (4)

2022 – 25

AIAA SciTech Forum (8)

2025 – 26

Transactions on Computers (1)

2023

Teaching

AERO 423: Orbital Mechanics

2024, 2025

Texas A&M University | *Teaching Assistant* [🔗](#)

Student feedback: "Your support and guidance during the courses have been incredibly helpful, and I truly appreciate you cheering me on as I prepare for my defense. It means a lot to have mentors like you."

Digital Signal Processing

2023

Texas A&M University | *Department Tutorial Lecture*

Mentoring

Ankit Burudgunte

2026

Rice University | *Undergraduate research (NSF REU), optical backscatter communication*

Dhawal Sirikonda

2026

Dartmouth College | *PhD project, digital signal processing*

Nevindu Batagoda

2026

Dartmouth College | *PhD project, event sensors*

Peiran Li

2026

Dartmouth College | *MS research, optics*

Omkar Vengurlekar Arizona State University <i>PhD project, audio signal processing</i>	2026
Team Lunatix PI for Lunar Autonomy Challenge 2025 <i>2 graduate and 3 undergraduate students, visual navigation on Moon</i>	2025
Omar Mohmand & Marco Peredo Texas A&M University <i>Undergraduate research: Trajectory design for Mars/Europa rendezvous</i>	2024

Volunteering

Dartmouth College Annual Science Day <i>Demonstrations to K-12 students</i>	2026
Montshire Museum of Science - Vermont Astronomy Day <i>Speaker, Astronomer in the House</i>	2026
Texas A&M University Aerospace Engineering Graduate Student Association (AEGSA) <i>Professional development chair</i> ↗	2023 – 2025
Texas A&M University Science and Astronomy Festival <i>Volunteer, outreach demonstrations</i>	2022 – 2025
Bhumi NGO Renuka High School, Bangalore <i>Volunteering physics teacher</i>	2017 – 2019

Software

GUISS: Graphical Utility for Icy Moon Surface Simulations ↗	2024
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